

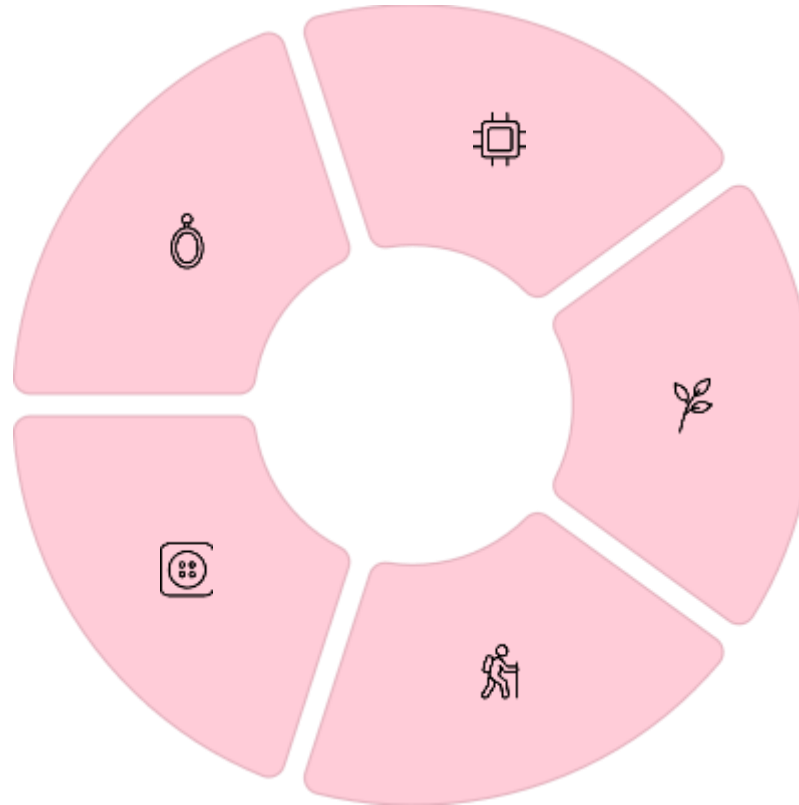
Attack Surface Categories

Page Table

Exploits side effects of page walking to observe memory access patterns at page granularity.

Enclave Interface

Analyzes ECALL/OCALL invocation patterns to infer sensitive input parameters.



CPU Cache

Leverages timing differences between cache hits and misses to infer execution patterns.

Branch Prediction

Abuses BTB, PHT, and RSB to extract control flow information at instruction level.

DRAM

Exploits row buffer timing to observe memory access patterns across virtual machines.

CPU Cache Vulnerabilities

- CPU caches bridge the speed gap between processors and DRAM, storing recently accessed data for fast retrieval. However, shared caches create timing channels that leak sensitive information about enclave execution.
- Cache controllers check whether requested data is cached (cache hit) or must be fetched from DRAM (cache miss). The **measurable time difference between hits and misses** enables attackers to infer which data the victim accessed, revealing secrets through careful timing analysis.

Prime+Probe Attack

■ How It Works

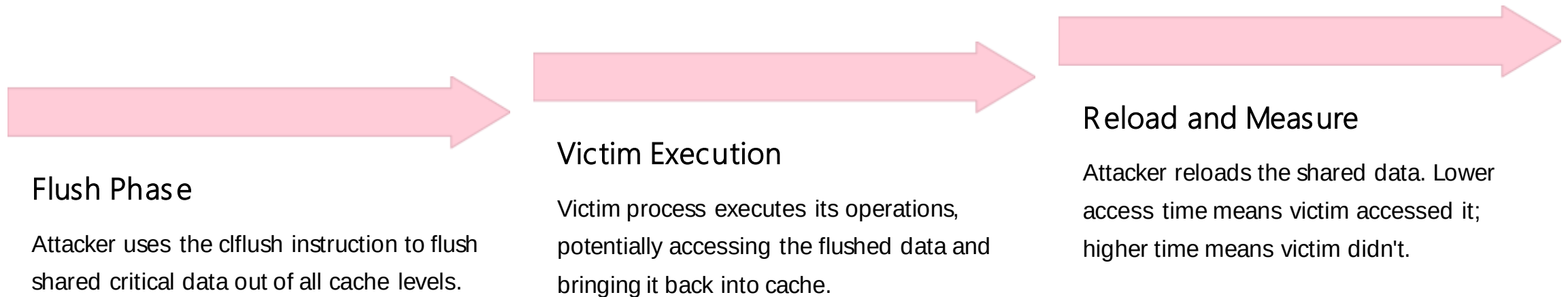
- **Initialization:** Attacker fills targeted cache sets with their own data, "priming" the cache.
- **Waiting:** Victim process executes, potentially evicting attacker's data from cache sets.
- **Measurement:** Attacker reloads their data and measures access time. Slower access indicates victim used that cache set.

■ Requirements

- Knowledge of LLC slice mapping
- Ability to create eviction sets
- No shared memory needed
- Works in cross-core scenarios

Flush+Reload Attack

Flush+Reload is one of the most powerful cache attack techniques, requiring shared memory between attacker and victim but providing high precision and low noise.



Additional Attack Techniques



Evict+Time

Attacker measures victim's total execution time after evicting specific cache lines. Higher execution time indicates evicted lines were accessed.



Flush+Flush

Measures the execution time of the clflush instruction itself. Fast execution means data was already flushed; slow means it was in cache.



Evict+Reload

Combines eviction with reload measurement. Attacker evicts data, victim executes, then attacker checks if their data was replaced.

Same-Core vs. Cross-Core Attacks

Same-Core Attacks

Victim and attacker processes run on the same processing core, accessing the shared L1 cache. Since L1 caches are virtually indexed, attackers don't need to understand virtual-to-physical address mapping.

These attacks are simpler but require the attacker to be scheduled on the same core as the victim.

Cross-Core Attacks

Victim and attacker run on different cores, targeting the shared Last Level Cache (LLC). These attacks require knowledge of virtual-to-physical address mapping and LLC slice mapping functions.

Cross-core attacks are more challenging but more realistic in multi-tenant cloud environments.

Attacker Knowledge



Trace-Driven Attacks

Attacker knows which cache lines the victim accessed and the exact order of these accesses. This provides the most detailed information but is hardest to obtain.



Access-Driven Attacks

Attacker has snapshots of cache state at various instances. Can infer which cache lines were accessed but not their order. Most practical attack category.



Time-Driven Attacks

Attacker analyzes variations in execution time of security-sensitive applications. Different secrets cause different numbers of cache hits/misses, leading to timing variations.

Cache Coherence Protocol Attacks

- Coherence protocols maintain data consistency across cache levels in multi-core systems. Most commercial processors use MESI or MOESI protocols.
- These protocols create vulnerabilities because read-access latency differs for blocks in different coherent states.
- Attacker may change coherent state of blocks to alter access times and transfer information.

DRAM-Based Attacks

- DRAM attacks exploit **timing differences in row buffer accesses** without requiring memory sharing between attacker and victim. The row buffer holds the most recently accessed row, enabling faster access than other memory locations.
- While DRAM attacks achieve near Flush+Reload accuracy without shared memory, they face limitations: **most memory resides in cache** rather than DRAM, accuracy reaches only 1KB granularity (4 DRAM rows per 4KB page), and **high noise levels** from multiple pages sharing DRAM rows. DRAM attacks are typically combined with CPU cache attacks for improved effectiveness.